CFormer: Cross-Frequency Transformer for Remote Sensing Semantic Segmentation

Yong Wang, Yuanyuan Li, Renzhe Wu, Wensong Zhang, Li An, and Guohua Geng

***Abstract*—Remote sensing semantic segmentation aims at fine-grained pixel-level understanding of high-resolution imagery, but remains challenging due to large-scale variations, complex textures, and uneven spatial distributions. Existing convolutional neural network (CNN) and transformer-based methods struggle to jointly model global context and local details. To address this issue, we propose a cross-frequency transformer for remote sensing semantic segmentation. The framework introduces a multiscale transformer with frequency-aware channel attention, which explicitly models low-frequency structural information and high-frequency details via discrete cosine transform (DCT), and adaptively fuses frequency-enhanced features to improve boundary preservation and cross-scale discrimination. Furthermore, a Fourier relation transformer is designed in the decoding stage to compute attention in the frequency domain, effectively strengthening long-range dependencies while suppressing high-frequency noise. Extensive experiments on six benchmark datasets demonstrate that the proposed method consistently outperforms existing approaches in overall accuracy (OA), boundary quality, and cross-scale object recognition.**

***Index Terms*—Cross-domain multiscale transformer, Fourier relational transformer (FRT), remote sensing imagery, semantic segmentation.**

I. Introduction

SEMANTIC segmentation of remote sensing imagery, a core task in Earth observation and geospatial information analysis, aims to perform fine-grained, pixel-level categorization of land-cover classes in high-resolution remote sensing images. With continuous advances in UAV, satellite, and aerial imaging technologies, remote sensing data now feature higher resolution, wider coverage, more diverse object classes, and more complex spatial structures. However, remote sensing images exhibit significant variations in object scales, highly diverse texture patterns, and irregular spatial layouts. Due to their reliance on local receptive fields, traditional convolutional neural networks (CNNs) struggle to effectively model multiscale spatial structures and long-range dependencies [1], [2], [3], [4], [5]. Consequently, the key challenge for improving semantic segmentation performance lies in simultaneously capturing cross-scale context, boundary details, and global structural information in remote sensing scenes.

CNNs and transformer architectures have become the two most important methodological paradigms for remote sensing semantic segmentation [6], [7], [8], [9], [10]. CNNs, leveraging local receptive fields and weight sharing, excel at extracting texture details and local structures, but their limited ability to model long-range relationships makes it difficult to capture the holistic structure of large-scale objects. In contrast, Transformer, with its self-attention mechanism, can explicitly model global relationships and demonstrate strong performance in describing object shapes and modeling cross-region consistency. However, existing transformer-based methods still have the following limitations: although self-attention captures long-distance structural relations, it remains insufficient for modeling the large-scale variations, complex texture details, and intraclass morphological diversity present in high-resolution remote sensing imagery. As a result, Transformer can still suffer from blurred boundaries, object adhesion, or class confusion when dealing with complex backgrounds, repetitive textures, or structures with cross-region consistency.

Recent research has attempted to enhance remote sensing semantic segmentation via multiscale structural modeling or frequency-domain information [11], [12], [13], [14], [15]. For instance, multiscale convolutions, dilated convolutions, and multiscale feature fusion techniques can improve the recognition of objects at varying scales; frequency-domain modeling methods based on Fourier transforms or wavelet features have also shown significant value in remote sensing image compression, enhancement, and recognition. Meanwhile, frequency-domain representations themselves offer strong advantages for remote sensing feature characterization. Although some existing approaches embed discrete cosine transform (DCT)/FFT operations into convolutional layers as auxiliary features, they do not establish a unified deep frequency-domain modeling framework and fail to fully exploit the complementarity of different frequency bands. Furthermore, integrating spatial and frequency-domain signals within a transformer—so that the model can simultaneously capture global structural shapes, enhance sensitivity to texture details, and maintain efficient cross-frequency and cross-scale feature interaction—remains a major research challenge.

To address these challenges, this article proposes a cross-frequency transformer framework for remote sensing semantic segmentation. The model adopts a U-shaped architecture, using a CNN-based encoder and transformer-based decoder, and explicitly combines the strengths of the spatial and

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frequency domains to jointly model multiscale structures, complex texture patterns, and long-range dependencies in remote sensing imagery.

The main contributions of this article are summarized as follows.

1) *Cross-Frequency Transformer:* This integrates multiscale spatial modeling with frequency-domain attention to enhance feature representation in complex remote sensing scenes.
2) *Cross-Frequency Multiscale Transformer:* This achieves joint spatial-frequency modeling through multiscale convolutions and DCT-based multifrequency channel attention, effectively improving boundary detail preservation and cross-scale object recognition.
3) *Fourier Relational Transformer:* This migrates self-attention into the 2-D frequency domain, using 2-D FFT to construct frequency-domain attention for efficient modeling of long-range dependencies and cross-region consistency in remote sensing images.

II. Related Work

A. CNN-Based Methods

CNNs were the earliest mainstream framework for remote sensing semantic segmentation. Relying on local convolutional kernels and hierarchical stacking, CNNs extract spatial features and aggregate contextual information through pyramid pooling, multiscale convolutions, or encoder–decoder structures. CNNs excel at modeling local textures and preserving spatial details, but their inherent limitation in receptive field size makes it difficult to capture large-range structural dependencies, especially in large-scale object recognition or complex scene understanding tasks [16], [17], [18].

To address the difficulty that a single architecture, either transformer or CNN, often struggles to simultaneously model both global and local features in remote sensing semantic segmentation, Yao et al. [19] proposed a hybrid architecture that combines Transformer and CNN. Their model employs a feature fusion module to capture dependencies across spatial positions and channels, enabling comprehensive multiscale feature integration and improving discrimination among visually similar classes such as building roofs and roads. In addition, a feature injection module compresses the transformer-extracted global information and injects it into the CNN branch, allowing the network to retain local sensitivity while incorporating richer global context. For fine-grained land-use/land-cover semantic segmentation in high-resolution remote sensing imagery, Zhang et al. [20] proposed a flexible multimodal segmentation framework based on the segment anything model. A dynamic multimodal feature fusion unit adaptively integrates multisource information, enhancing cross-modal complementarity; a dynamic attention mechanism strengthens contextual understanding across modalities; and a semantic cross-modal integration module alleviates semantic misalignment and feature inconsistency between modalities, ensuring coherent and discriminative fused representations.

B. Transformer-Based Methods

Transformer, equipped with self-attention mechanisms, explicitly captures global dependencies and has been widely applied in remote sensing semantic segmentation. Their advantages include the ability to model long-range relationships and large-scale structures. However, they also suffer from high computational cost, sensitivity to noise, and limited ability to capture high-frequency textures [21], [22], [23], [24], [25].

To address issues such as large-scale variations and significant intraclass appearance differences in high-resolution remote sensing imagery, Zeng et al. [26] proposed a semantic segmentation network capable of efficiently modeling multiscale context and global dependencies. Their approach uses a cross-attention mechanism during encoding to promote interaction among features at different scales, effectively bridging the semantic gap between low and high-level features and enriching feature representations. Furthermore, a windowed efficient multihead self-attention mechanism models long-range dependencies in both spatial and channel dimensions while enabling cross-window interaction, balancing global context capture with computational efficiency.

To address challenges such as complex backgrounds, significant scale and orientation variations, and large intraclass diversity in remote sensing segmentation, Ma et al. [27] proposed an adaptive class-aware network integrating both global and local semantic perception. A global class-aware module extracts class-level contextual features and captures global interclass dependencies, mitigating background noise and enhancing semantic consistency. A local class-aware module generates localized class representations as intermediate semantic cues, indirectly associating pixels with global class semantics and reducing intraclass variation and texture diversity issues.

C. Frequency Domain-Based Methods

In recent years, frequency-domain approaches have received increasing attention. By mapping images into the frequency domain via Fourier or DCTs, they disentangle low-frequency components (global structures) from high-frequency components (textures and details). This makes frequency-domain methods naturally suitable for denoising, edge enhancement, and texture modeling [28], [29], [30], [31].

To address boundary uncertainty caused by drastic grayscale variations and overlapping categories in remote sensing images, Wei et al. [32] proposed a spatial-frequency dual-domain fusion network. A multilevel wavelet decomposition module extracts high-frequency and low-frequency features, enhancing the separability of categories with similar spatial patterns. A fuzzy spatial constraint module uses higher order fuzzy modeling with upper and lower membership functions to adaptively constrain boundary features, alleviating boundary ambiguity. A dual-domain feature fusion module aligns and integrates frequency and spatial domain features to achieve cooperative modeling and improve segmentation accuracy. Zhang et al. [33] proposed a frequency-domain guided feature coupled network to address the limitations of existing methods in modeling frequency-domain information, which often fail to effectively capture fine details and thus perform poorly in complex scenes. This method enhances the representation of frequency and spatial domain features, as well as local and global features, through a Swin transformer architecture guided by frequency-domain information. By incorporating dilated group convolutions, fast Fourier

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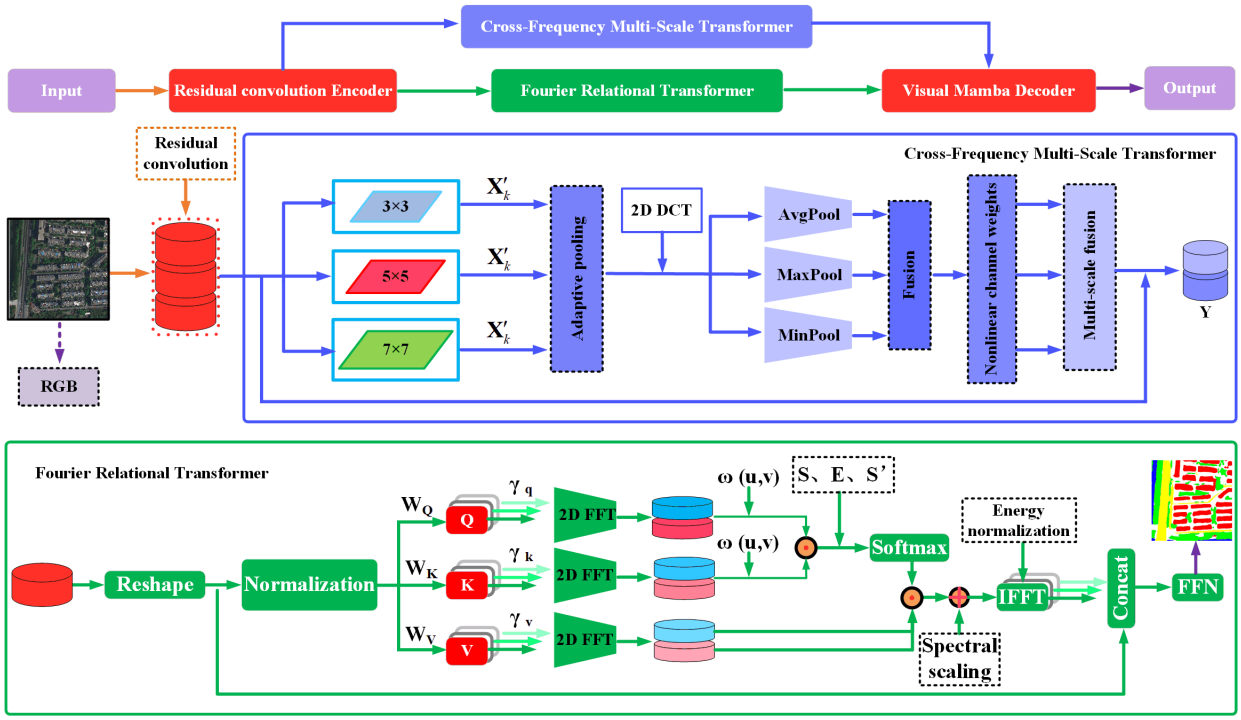


Fig. 1. Network architecture of the overall.

transform, and learnable weights, the proposed algorithm effectively fuses frequency and spatial-domain information, thereby improving the ability to capture details in remote sensing images—particularly in regions with complex textures and fine structures.

Beyond semantic segmentation, remote sensing interpretation has recently advanced toward more generalizable models across modalities and tasks, such as foundation-style hyperspectral video tracking [34]. Meanwhile, state-space modeling has gained attention in SAR understanding, where MaDiNet integrates Mamba with a diffusion formulation for SAR target detection [35]. A latest survey further summarizes 50 years of SAR ATR and outlines future directions toward more generalizable and physically consistent SAR perception [36]. These trends motivate exploring unified and efficient representation learning strategies for remote sensing imagery.

Given the challenges faced by CNN, transformer, and frequency domain-based semantic segmentation methods in different scenarios, this article proposes a cross-frequency transformer for remote sensing semantic segmentation.

III. Method

A. Overall Framework

The proposed CFormer adopts a U-shaped encoder–decoder architecture and integrates frequency-domain enhancement into both the bottleneck and skip-connection pathways, enabling joint spatial-frequency feature modeling. The framework consists of four main components: a residual convolution encoder, the cross-frequency multiscale transformer (CFMST) for skip connections, the Fourier relational transformer (FRT) at the bottleneck, and a Visual Mamba decoder. The overall architecture is illustrated in Fig. 1.

Unlike FcaNet's [37] frequency-based channel compression, our approach leverages DCT components for cross-frequency token interaction rather than channel-wise recalibration.

The encoder progressively extracts hierarchical spatial representations, ranging from local textures to high-level semantic features, providing a stable foundation for subsequent frequency-domain modeling. At the bottleneck, the FRT projects features into the 2-D frequency domain and constructs relational attention via complex-domain interactions, enabling compact modeling of long-range dependencies and global structural consistency.

In parallel, skip-connection features are refined by the CFMST, which employs multiscale convolutional branches and multifrequency channel attention to jointly capture cross-scale variations and cross-frequency structural cues, including low-frequency global structures and high-frequency boundary details.

During decoding, the FRT-enhanced bottleneck features are progressively upsampled and fused with CFMST-refined skip features. The visual Mamba module then performs state-space-based contextual refinement to further enhance long-range consistency and spatial detail recovery, ultimately producing accurate pixel-wise segmentation outputs.

B. Cross-Frequency Multiscale Transformer

Remote sensing images exhibit pronounced cross-scale variations (e.g., large differences among houses, roads, and water bodies) and rich fine-grained textures (e.g., farmland patterns and building edges). Traditional convolution and channel-attention methods often operate at a single scale or rely on a single type of statistical descriptor, yielding insufficient expressive power for complex remote sensing scenes. To address this, we propose the CFMST, which achieves

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cross-scale and cross-frequency modeling through geometric multiscale convolution representations and multifrequency channel attention.

Multiscale spatial features are first extracted through parallel convolutional branches with different receptive fields. To incorporate spectral structural cues, we introduce a DCT-based multifrequency channel attention mechanism. A predefined set of 2-D frequency components is selected using a deterministic indexing strategy, and the corresponding DCT bases are analytically constructed and kept fixed during training. Only the channel reweighting function applied to the aggregated frequency responses is learnable. This design maintains the interpretability of frequency decomposition while avoiding additional learnable frequency-location parameters that may cause instability or overfitting. By combining multiscale representations with fixed cross-frequency bases and adaptive channel reweighting, CFMST enhances structural and texture-aware feature modeling for remote sensing segmentation.

1) Multiscale Convolutional Geometric Representation: Given input features $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$ (B is the batch size, C is the number of channels, and (H, W) are the spatial size), define a set of convolution kernels at three different scales

$$\mathbf{X}_k = \phi\left(\mathrm{BN}\left(W_k * \mathbf{X}\right)\right), \quad k \in 3, 5, 7 \tag{1}$$

where W_k denotes a $k \times k$ convolution kernel, and ϕ is the activation function.

Furthermore, we introduce a spatial expansion factor and scale normalization to enhance stability

$$\mathbf{X}'_k = \frac{\mathbf{X}_k}{\sqrt{\frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathbf{X}_k\left(i,j\right)^2 + \epsilon}}. \tag{2}$$

2) Multifrequency Attention: To capture the differences between high-frequency and low-frequency information in remote sensing images (e.g., high frequencies correspond to edges and textures, while low frequencies correspond to semantic contours), we construct a channel attention mechanism based on the DCT.

Adaptive Pooling and Frequency Block Mapping: To map inputs of different sizes into the same frequency domain, we design a linear–nonlinear hybrid mapping $\mathbb{R}^{C\times H\times W} \rightarrow \mathbb{R}^{C\times h_0\times w_0}$

$$\widetilde{\mathbf{X}}_k = \mathrm{AdaptiveAvgPool}_{(h_0,w_0)}\left(\mathbf{X'}_k\right). \tag{3}$$

Construction of 2-D DCT Basis Functions: The 2-D DCT basis functions are defined as

$$F_{u,v}\left(x,y\right) = \alpha\left(u\right)\alpha\left(v\right)\cos\left[\frac{\left(2x+1\right)u\pi}{2h_0}\right]\cos\left[\frac{\left(2y+1\right)v\pi}{2w_0}\right] \tag{4}$$

where the normalization coefficient is

$$\alpha\left(t\right) = \begin{cases}\sqrt{\dfrac{1}{n}}, & t = 0\\ \sqrt{\dfrac{2}{n}}, & t > 0.\end{cases} \tag{5}$$

A set of basis functions is selected from all possible frequency combinations

$$\mathcal{F} = \left(u_1, v_1\right), \left(u_2, v_2\right), \ldots, \left(u_K, v_K\right). \tag{6}$$

Multifrequency Response Statistics: For each frequency pair (u, v), the frequency response is defined as

$$\mathbf{R}_{u,v}\left(c\right) = \sum_{x=0}^{h_0-1}\sum_{y=0}^{w_0-1}\widetilde{\mathbf{X}}_k\left(c,x,y\right)F_{u,v}\left(x,y\right). \tag{7}$$

To enhance the diversity of the statistics, three types of pooling are introduced

$$A_{u,v}\left(c\right) = \mathrm{AvgPool}\left(\mathbf{R}_{u,v}\left(c\right)\right) \tag{8}$$

$$M_{u,v}\left(c\right) = \mathrm{MaxPool}\left(\mathbf{R}_{u,v}\left(c\right)\right) \tag{9}$$

$$N_{u,v}\left(c\right) = \mathrm{MinPool}\left(\mathbf{R}_{u,v}\left(c\right)\right). \tag{10}$$

The aggregated frequency vector is

$$Z\left(c\right) = \frac{1}{K}\sum_{(u,v)\in\mathcal{F}}\left(A_{u,v}\left(c\right) + M_{u,v}\left(c\right) + N_{u,v}\left(c\right)\right). \tag{11}$$

Nonlinear Channel Weights: The channel weights are generated by

$$W\left(c\right) = \sigma\left(W_2, \delta\left(W_1 Z\left(c\right)\right)\right) \tag{12}$$

where, W_1 and W_2 are linear layers.

Final Output:

$$\mathbf{X}_k^{\mathrm{MFA}}\left(c\right) = W\left(c\right)\cdot\mathbf{X}_k\left(c\right). \tag{13}$$

3) Multiscale Fusion: Three-scale concatenation

$$\mathbf{X}_{\mathrm{cat}} = \mathrm{Concat}\left[\mathbf{X}_3^{\mathrm{MFA}}, \mathbf{X}_5^{\mathrm{MFA}}, \mathbf{X}_7^{\mathrm{MFA}}\right]. \tag{14}$$

Fusion using a 1×1 convolution

$$\mathbf{Y}' = \phi\left(\mathrm{BN}\left(W_f * \mathbf{X}_{\mathrm{cat}}\right)\right). \tag{15}$$

Residual enhancement is introduced

$$\mathbf{Y} = \mathrm{Conv}_{3\times3}\left(\mathbf{Y}' + \mathbf{X}\right). \tag{16}$$

C. Fourier Relational Transformer

The attention mechanism in transformer fundamentally relies on the dot-product structure of features in the spatial domain. However, repetitive textures commonly found in remote sensing images—such as roof tiles or road patterns—exhibit distinct separability in the frequency domain. Therefore, modeling solely in the spatial domain may ignore important frequency structures of the features. To address this, we propose the FRT module, which fully transfers the traditional attention mechanism into the 2-D frequency domain. It incorporates complex-valued computations, frequency-domain weight normalization, spectral scaling factors, and spectral energy enhancement terms to more comprehensively model long-range dependencies and frequency responses in remote sensing imagery.

The overall transformation pipeline of FRT can be summarized as: sequence-to-spatial reshaping → channel normalization → multihead projection → frequency-domain transformation → complex-valued interaction → inverse frequency projection → spatial-domain normalization → residual fusion. Explicit dimension tracking is maintained at each stage to ensure reproducibility and eliminate ambiguity in tensor transitions.

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1) Spatial-to-Sequence Transformation: For the encoder output sequence $\mathbf{X} \in \mathbb{R}^{B\times N\times C}(N = H \cdot W)$, it is reshaped from a sequence back into a 2-D feature tensor

$$\mathbf{X}_s = \text{reshape}\,(\mathbf{X}) \in \mathbb{R}^{B\times C\times H\times W}. \tag{17}$$

To ensure numerical stability in the frequency-domain transformation, normalization is further applied

$$\widehat{\mathbf{X}}_s = \frac{\mathbf{X}_s - \mu\,(\mathbf{X}_s)}{\sqrt{\sigma^2\,(\mathbf{X}_s) + \epsilon}} \tag{18}$$

where μ and σ denote the mean and standard deviation computed per channel, respectively.

A 1×1 convolution is used to generate

$$\mathbf{Q} = W_q * \widehat{\mathbf{X}}_s, \quad \mathbf{K} = W_k * \widehat{\mathbf{X}}_s, \quad \mathbf{V} = W_v * \widehat{\mathbf{X}}_s. \tag{19}$$

It is then split into multiple heads

$$\mathbf{Q}, \mathbf{K}, \mathbf{V} \in {}^{B\times h\times d\times H\times W}, \quad C = hd \tag{20}$$

where B denotes batch size, h is the number of attention heads, d is the per-head channel dimension, and $C = hd$ ensures dimension consistency between the projected multihead representation and the original channel space.

To enhance frequency-domain stability, a scalar scaling factor is introduced

$$\mathbf{Q}' = \gamma_q\mathbf{Q}, \quad \mathbf{K}' = \gamma_k\mathbf{K}, \quad \mathbf{V}' = \gamma_v\mathbf{V} \tag{21}$$

where $\gamma_q, \gamma_k, \gamma_v \in \mathbb{R}$ are learnable parameters that improve the model's ability to adjust frequency responses.

2) 2-D FFT: A 2-D FFT is applied to each head

$$\mathbf{Q}_f = \mathcal{F}_{2D}\left(\mathbf{Q}'\right), \quad \mathbf{K}_f = \mathcal{F}_{2D}\left(\mathbf{K}'\right), \quad \mathbf{V}_f = \mathcal{F}_{2D}\left(\mathbf{V}'\right). \tag{22}$$

After the 2-D FFT, the transformed features become $Q_f, K_f, V_f \in \mathbb{C}^{B\times h\times d\times H\times W}$. The real and imaginary components are then separated as $Q_r, Q_i \in \mathbb{R}^{B\times h\times d\times H\times W}$, ensuring explicit dimension tracking throughout the frequency interaction process.

The complex features are then split into real and imaginary parts

$$\mathbf{Q}_f = \mathbf{Q}_r + j\mathbf{Q}_i, \quad \mathbf{K}_f = \mathbf{K}_r + j\mathbf{K}_i, \quad \mathbf{V}_f = \mathbf{V}_r + j\mathbf{V}_i. \tag{23}$$

Frequency-axis coordinates are defined as

$$u \in [0, H-1], \quad v \in [0, W-1] \tag{24}$$

where the indices (u, v) correspond to discrete frequency coordinates in the 2-D Fourier spectrum, where lower values represent global structural components and higher values capture fine-grained spatial variations.

To enhance low-frequency feature stability, an additional frequency decay factor is introduced

$$\omega\,(u, v) = \left(1 + \frac{u^2 + v^2}{H^2 + W^2}\right)^{-\beta}, \quad \beta > 0. \tag{25}$$

The decay factor $\omega(u, v)$ assigns larger weights to low-frequency components while gradually attenuating high-frequency amplitudes, preventing excessive spectral amplification and improving numerical stability.

Frequency weights are then used to modulate Q and K

$$\widetilde{\mathbf{Q}}_f\,(u, v) = \omega\,(u, v)\,\mathbf{Q}_f\,(u, v), \quad \widetilde{\mathbf{K}}_f\,(u, v) = \omega\,(u, v)\,\mathbf{K}_f\,(u, v). \tag{26}$$

TABLE I
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE ISPRS VAIHINGEN DATASET

Method	Surf.	Building	LowVeg.	Tree	Car	mIoU	mF1	OA
ABCNet	92.5	95.4	83.9	89.6	85.4	81.1	89.3	90.4
Dcswin	93.6	96.1	85.7	90.3	87.6	83.2	90.6	91.6
Unetformer	92.5	95.5	83.7	89.5	88.2	81.9	89.8	90.6
ConvlsrNet	96.6	95.0	84.2	89.8	87.2	83.1	90.5	92.9
MIEFNet	93.3	95.9	85.5	90.3	90.6	83.9	91.1	91.4
MCKTNet	93.2	96.1	85.1	90.4	89.1	83.4	90.7	91.4
FDMF-Net	-	-	-	-	-	83.8	90.9	91.9
AFENet	96.9	95.7	85.0	90.6	89.3	84.5	91.5	91.6
UrbanSSF	96.7	95.6	84.3	89.7	88.9	84.1	91.0	93.0
Ours	96.9	96.1	84.8	90.0	89.8	84.7	91.5	93.7

3) Frequency-Domain Attention: The complex dot product is defined as

$$\widetilde{\mathbf{Q}}_f \cdot \widetilde{\mathbf{K}}_f^{\dagger} = \left(\mathbf{Q}_r\mathbf{K}_r^{\top} + \mathbf{Q}_i\mathbf{K}_i^{\top}\right) + j\left(\mathbf{Q}_i\mathbf{K}_r^{\top} - \mathbf{Q}_r\mathbf{K}_i^{\top}\right) \tag{27}$$

where † denotes the complex conjugate transpose

$$\mathbf{S} = \text{Re}\left(\widetilde{\mathbf{Q}}_f\widetilde{\mathbf{K}}_f^{\dagger}\right) = \widetilde{\mathbf{Q}}_r\widetilde{\mathbf{K}}_r^{\top} + \widetilde{\mathbf{Q}}_i\widetilde{\mathbf{K}}_i^{\top}. \tag{28}$$

To enhance frequency-domain energy representation, a spectral energy enhancement term is introduced

$$\mathbf{E} = \eta \cdot \left(\left|\widetilde{\mathbf{Q}}_f\right| \cdot \left|\widetilde{\mathbf{K}}_f\right|^{\top}\right), \quad |\cdot| = \sqrt{(\cdot)_r^2 + (\cdot)_i^2}. \tag{29}$$

The final score matrix is

$$\mathbf{S}' = \frac{\mathbf{S} + \mathbf{E}}{\sqrt{d}}. \tag{30}$$

Attention weights

$$\mathbf{A} = \text{softmax}\left(\mathbf{S}'\right). \tag{31}$$

The output is then computed as

$$\mathbf{O}_r = \mathbf{A}\mathbf{V}_r, \quad \mathbf{O}_i = \mathbf{A}\mathbf{V}_i. \tag{32}$$

Fused as

$$\mathbf{O}_f = \mathbf{O}_r + j\mathbf{O}_i. \tag{33}$$

To control frequency-domain amplitudes, spectral scaling is applied

$$\widehat{\mathbf{O}}_f = \lambda\mathbf{O}_f, \quad \lambda \in \mathbb{R}. \tag{34}$$

4) Inverse FFT: The inverse FFT is applied

$$\mathbf{O}_s = \text{Re}\left(\mathcal{F}_{2D}^{-1}\left(\widehat{\mathbf{O}}_f\right)\right). \tag{35}$$

To avoid a numerical explosion caused by the inverse transform, energy normalization is additionally applied

$$\widetilde{\mathbf{O}}_s = \frac{\mathbf{O}_s}{\sqrt{\frac{1}{HW}\sum_{i,j}\mathbf{O}_s^2\,(i, j) + \epsilon}}. \tag{36}$$

5) Multihead Fusion: Heads are concatenated and linearly mapped

$$\mathbf{Y}_s = W_o * \text{Concat}\left(\widetilde{\mathbf{O}}_s\right) + \mathbf{X}_s. \tag{37}$$

After inverse FFT and normalization, each attention head produces spatial-domain features of size $\mathbb{R}^{B\times d\times H\times W}$, which are concatenated along the channel dimension to recover $\mathbb{R}^{B\times C\times H\times W}$.

After passing through the feedforward network, the final output is reshaped back into the sequence $\mathbf{Y} \in \mathbb{R}^{B\times N\times C}$

$$\mathbf{Y} = \text{FFN}\,(\mathbf{Y}_s) = W_2\phi\,(W_1\mathbf{Y}_s) + \mathbf{Y}_s. \tag{38}$$

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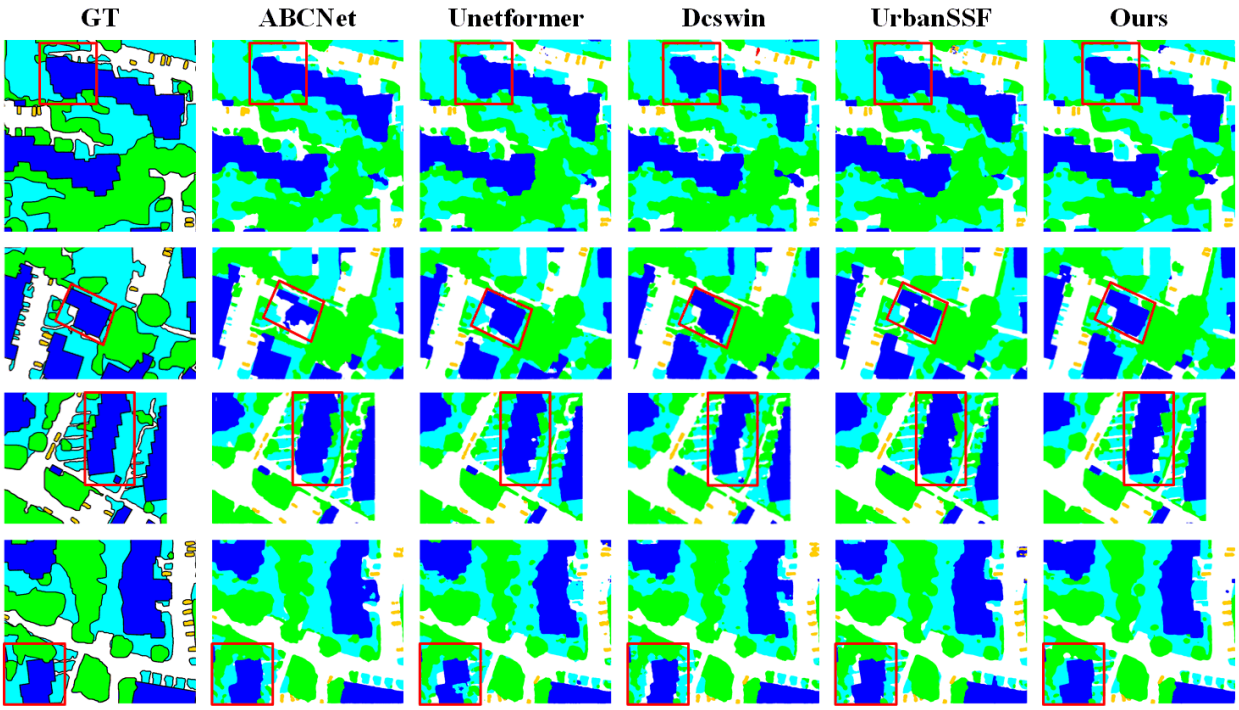


Fig. 2. Visualizations for the Vaihingen dataset.

TABLE II
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE ISPRS POTSDAM DATASET

Method	Surf.	Building	LowVeg.	Tree	Car	mIoU	mF1	OA
ABCNet	92.4	96.5	86.6	87.9	95.1	85.0	91.7	90.3
Dcswin	94.1	97.5	88.5	89.6	96.3	87.5	93.2	92.0
Unetformer	93.3	96.6	86.7	87.8	95.9	85.6	92.0	90.6
ConvlsrNet	94.0	96.9	87.8	88.8	96.0	86.7	92.7	91.5
MIEFNet	93.3	97.2	87.8	89.2	96.4	86.8	92.7	91.3
MCKTNet	93.2	97.1	87.9	89.1	96.3	86.7	92.7	91.2
FDMF-Net	-	-	-	-	-	86.4	92.5	91.5
AFENet	94.5	96.8	88.1	89.7	96.8	87.5	93.1	92.0
UrbanSSF	94.2	96.6	87.8	89.0	96.1	86.9	92.7	91.7
Ours	94.6	97.8	88.7	90.2	95.6	87.5	93.3	92.1

6) Computational Complexity Analysis: Let the spatial resolution be $H \times W$ and denote $N = HW$. In conventional spatial self-attention, the global dot-product operation $QK^\top$ requires $\mathcal{O}(N^2C)$ computational complexity due to the pairwise interaction between all spatial tokens, which becomes prohibitively expensive for high-resolution remote sensing imagery.

In contrast, the proposed FRT transfers the interaction into the frequency domain. The 2-D FFT applied to each feature map requires $\mathcal{O}(N \log N)$ operations. The subsequent frequency-domain interaction is performed element-wise across frequency coordinates, leading to $\mathcal{O}(NC)$ complexity. Finally, the inverse FFT also incurs $\mathcal{O}(N \log N)$ computational cost. Therefore, the overall complexity of the proposed FRT is $\mathcal{O}(N \log N + NC)$.

When N is large, as in high-resolution remote sensing scenarios, $N \log N$ is significantly smaller than N^2, indicating that the proposed formulation effectively reduces the quadratic spatial dependency inherent in global self-attention while preserving long-range modeling capacity.

7) Stability and Boundedness Analysis: Although the proposed FRT introduces complex-valued frequency interactions, it remains a single-layer feed-forward attention mapping followed by a residual connection, without any recursive or iterative spectral update. Therefore, its convergence behavior is consistent with that of standard Transformer blocks.

To ensure numerical stability, several bounded transformations are introduced. Channel-wise normalization is applied prior to FFT to stabilize input variance. Learnable scalar factors γ_q, γ_k, and γ_v regulate frequency amplitudes. The frequency decay function $\omega(u, v)$ suppresses excessive high-frequency amplification. After inverse FFT, energy normalization constrains the spatial-domain output magnitude

$$O_s^e = \frac{O_s}{\sqrt{\frac{1}{HW}\sum_{i,j} O_s^2(i,j) + \epsilon}} \tag{39}$$

which guarantees bounded feature responses. Consequently, all intermediate activations remain numerically controlled, ensuring stable gradient propagation.

IV. EXPERIMENTS

A. Dataset, Implementation Details, and Evaluation Metrics

1) Dataset: In our experiments, we evaluate performance on ISPRS Potsdam [38], ISPRS Vaihingen [39], iSAID [40], LoveDA [41], UAVid [42], and WHU-OPT-SAR [43]. These datasets cover diverse sensor types, spatial resolutions, scene characteristics, and task focuses, providing a comprehensive benchmarking platform.

a) ISPRS Vaihingen and Potsdam: ISPRS Vaihingen and Potsdam are high-resolution urban surface classification benchmarks focused on building, road, tree, and other land-cover classes, containing a total of six categories. All images are preprocessed and resized to 1024×1024 pixels.

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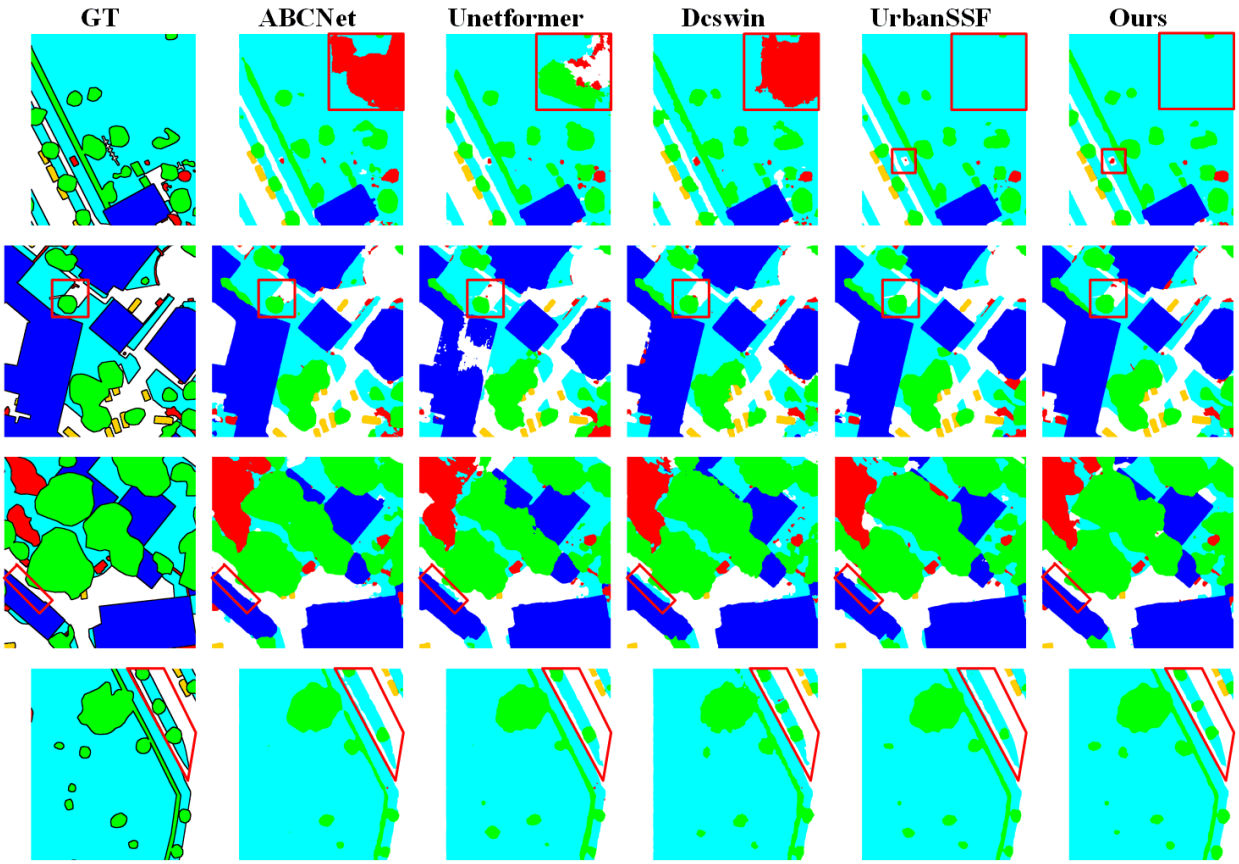


Fig. 3. Visualizations for the Potsdam dataset.

TABLE III
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE UAVID DATASET

Method	Clut.	Building	Road	Tree	LowVeg.	Mov.Car	Sta.Car	Human	mIoU	FLOPs(G)	FPS	Params(M)
ABCNet	67.4	86.4	81.2	79.9	63.1	69.8	48.4	13.9	63.8	63.2	101.2	13.4
Dcswin	52.4	83.5	81.0	80.1	71.7	58.3	52.4	24.3	62.9	140.2	48.9	66.9
Unetformer	64.4	85.0	78.7	79.1	62.7	69.9	60.0	28.6	66.1	47.1	110.9	11.7
Mamba-UNet	56.6	78.5	76.1	71.5	53.0	66.4	40.5	15.9	57.3	160.3	7.1	35.9
Swin-UMamba	52.3	76.8	73.9	71.0	50.3	61.6	26.1	15.4	53.4	657.5	6.8	59.8
VM-UNet	56.0	78.1	75.5	73.1	54.9	65.1	31.0	11.5	55.7	157.3	7.2	44.3
ConvlsrNet	59.0	83.5	80.7	80.5	62.0	68.6	65.6	41.4	67.6	142.6	42.0	68.1
UrbanSSF	68.3	88.1	80.7	80.8	64.4	73.9	69.2	31.8	69.8	448.2	10.2	60.0
Ours	69.2	93.0	81.6	80.7	72.7	77.7	75.3	52.2	75.3	426.2	23.1	63.8

b) iSAID: iSAID is a large-scale aerial imagery dataset for instance-level analysis, containing many objects with large-scale variations. It includes 16 classes (including background). After horizontal and vertical augmentation, the dataset consists of 8154 images: 6894 for training, 837 for validation, and 423 for testing. All images are resized to 800 × 800 pixels.

c) LoveDA: LoveDA contains complex urban–rural mixed scenes. It includes 5987 high-resolution remote sensing images from Nanjing, Changzhou, and Wuhan (1024 × 1024 pixels). The dataset spans both urban and rural environments, featuring multiscale objects, complex backgrounds, and inconsistent class distributions.

d) UAVid: UAVid is a UAV-view urban road-scene dataset characterized by wide parallax and complex textures. All images are preprocessed, cropped, and resized to 1024 × 1024 pixels.

e) WHU-OPT-SAR: WHU-OPT-SAR is an optical–SAR fusion dataset containing cross-modal noise and frequency-domain discrepancies. It covers diverse terrains such as mountains, forests, hills, and plains, as well as various vegetation types, including coniferous, broadleaf, shrub, and aquatic vegetation. All images are resized to 1024 × 1024 pixels.

2) Implementation Details: Our hyperparameter settings follow ABCNet [42] and Unetformer [43]. Training is conducted for 80 epochs, with a learning rate of $6e^{-4}$ and weight decay of 0.01. The model is trained on a system equipped with 4× Nvidia L40s GPUs, an Intel[1] Xeon[1] Gold 6530 processor, 1-TB RAM, running Ubuntu 22.04.5 LTS.

3) Evaluation Metrics: Following prior work, for the ISPRS Vaihingen, Potsdam, iSAID, and WHU-OPT-SAR datasets, we use overall accuracy (OA), mean intersection over

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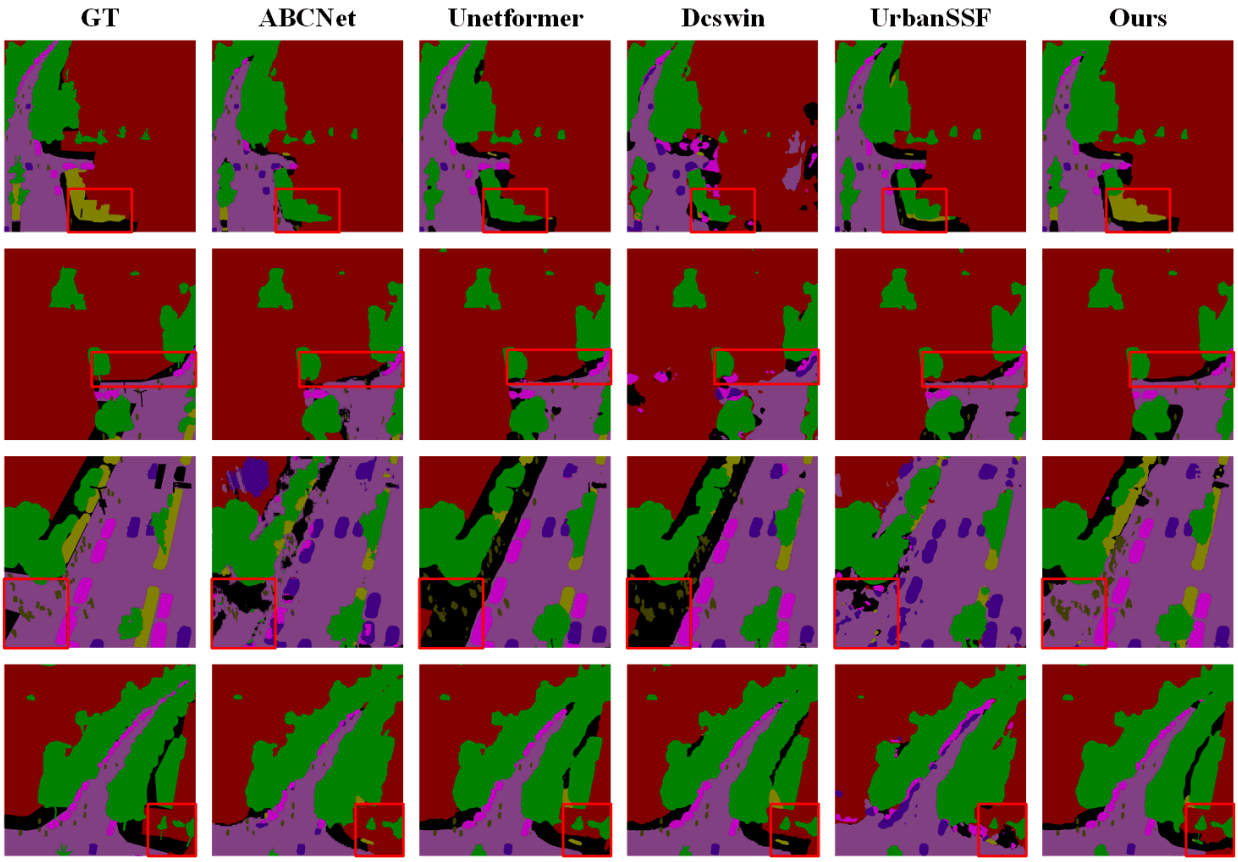


Fig. 4. Visualizations for the UAVid dataset.

TABLE IV
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE iSAID DATASET

Method	mIoU	mF1	OA
ABCNet	34.8	51.0	93.8
Dcswin	35.7	54.0	93.7
Unetformer	35.2	55.6	93.9
ConvlsrNet	34.8	52.6	94.6
UrbanSSF	39.0	54.2	93.9
Ours	44.4	61.7	94.1

union (mIoU), and $F1$-score ($F1$) as evaluation metrics. For the LoveDA and UAVid datasets, mIoU is used as the primary evaluation metric.

B. ISPRS Vaihingen and Potsdam Datasets

To validate the performance of CFormer on high-spatial resolution imagery, we conduct experiments on the ISPRS Vaihingen dataset. We also compare our method with state-of-the-art algorithms including ABCNet [44], Dcswin [45], Unetformer [46], MIEFNet [47], MCKTNet [48], ConvlsrNet [49], FDMF-Net [50], AFENet [15], and UrbanSSF [51].

From Table I and Fig. 2, it can be seen that the CFormer algorithm outperforms all other methods in the quantitative evaluation on the ISPRS Vaihingen dataset. In terms of m$F1$ comparison, CFormer achieves improvements of 0.5%, 1.7%, 0.8%, and 2.2% over UrbanSSF, Unetformer, Dcswin, and ABCNet, respectively.

To further verify the performance of the CFormer algorithm in complex urban building structures, we conducted training and validation on the ISPRS Potsdam dataset. In addition, we compared CFormer with other advanced algorithms on ISPRS Potsdam, including ABCNet, Dcswin, Unetformer, MIEFNet, MCKTNet, ConvlsrNet, and UrbanSSF.

From Table II and Fig. 3, it can be seen that the CFormer algorithm again outperforms all other methods in the quantitative evaluation on the ISPRS Potsdam dataset. In terms of m$F1$, CFormer achieves improvements of 0.6%, 0.6%, 1.3%, and 1.6% over UrbanSSF, ConvlsrNet, Unetformer, and ABCNet, respectively.

C. UAVid Dataset

To further verify the performance of the CFormer algorithm under dynamic, non-nadir imaging conditions, we conducted experiments on the UAVid dataset. We also compared CFormer with other advanced algorithms on UAVid, including ABCNet, Dcswin, Unetformer, Mamba-UNet, Swin-UMamba, VM-UNet, ConvlsrNet, and UrbanSSF.

From Table III and Fig. 4, it can be seen that the CFormer algorithm outperforms all other methods in the quantitative evaluation on the UAVid dataset. In terms of mIoU comparison, CFormer achieves improvements of 5.5%, 7.7%, 12.4%, 18%, and 11.5% over UrbanSSF, ConvlsrNet, Mamba-UNet, Unetformer, and ABCNet, respectively. In addition to accuracy, we analyze computational complexity. CFormer maintains a competitive parameter size, comparable

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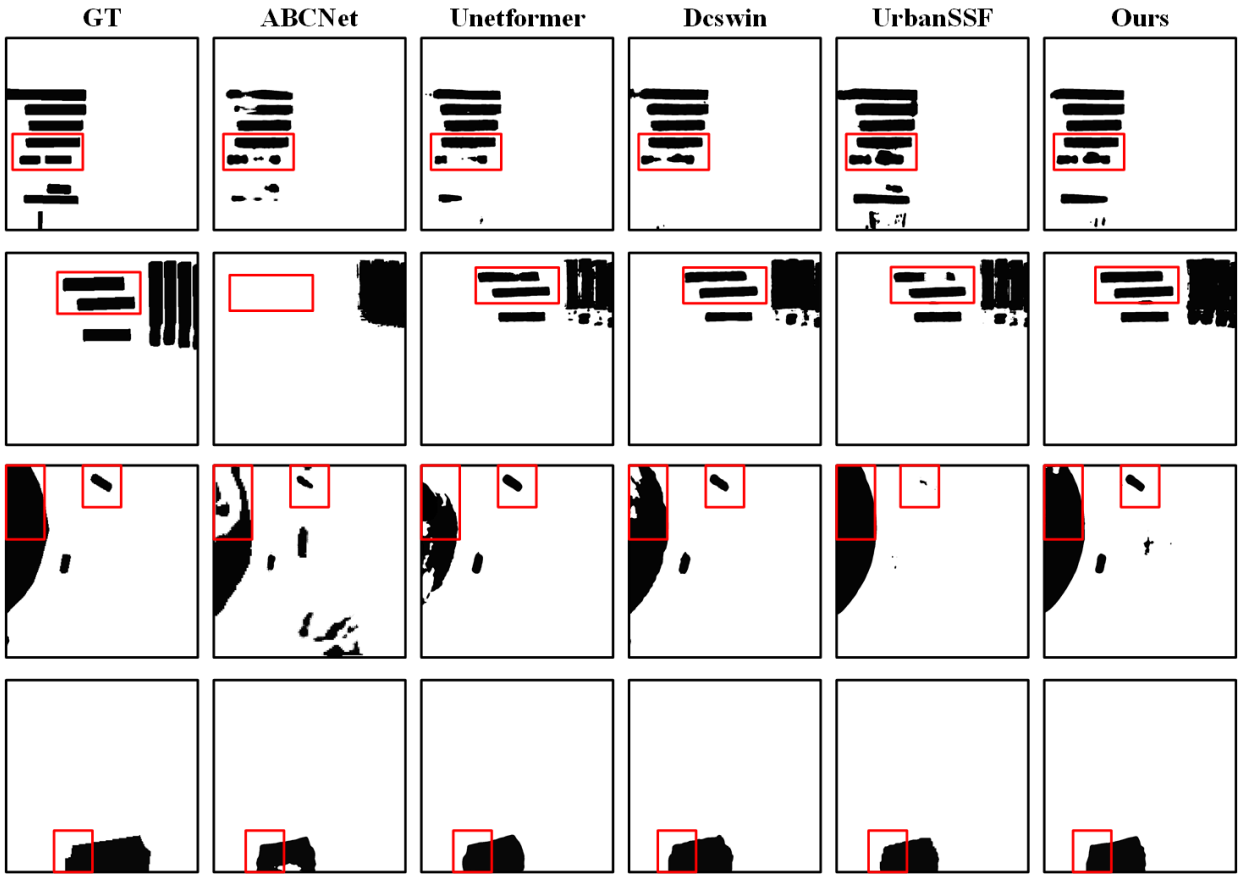


Fig. 5. Visualizations for the iSAID dataset.

TABLE V
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE LOVEDA DATASET

Method	Back.	Build.	Road	Water	Bar.	Forest	Agr.	mIoU
ABCNet	52.7	63.9	53.3	59.7	30.4	41.9	49.5	50.2
Dcswin	41.3	54.5	56.2	78.1	14.5	47.2	62.4	50.6
Unetformer	44.7	58.8	54.9	79.6	20.1	46.0	62.4	52.4
ConvlsrNet	47.0	62.6	57.1	82.0	21.3	48.5	64.6	54.7
SegNeXt	46.0	61.4	53.9	81.5	19.9	48.5	65.9	53.9
AFENet	47.4	59.2	59.1	81.5	21.3	48.6	66.4	54.7
UrbanSSF	51.6	62.8	56.0	62.0	39.2	40.7	48.2	51.5
Ours	55.6	68.1	56.3	72.0	34.8	44.9	57.7	55.6

to Dcswin and UrbanSSF. Although its FLOPs exceed those of lightweight models such as Unetformer, the accuracy gain is substantial. Moreover, CFormer requires fewer FLOPs than Swin-UMamba while delivering superior performance. With an inference speed of 23.1 FPS, CFormer achieves a favorable balance between segmentation accuracy and computational efficiency.

D. ISAID Dataset

To further evaluate the fine-grained segmentation capability of CFormer on multicategory targets—particularly small and densely distributed objects. In complex scenes, we conducted validation experiments on the large-scale aerial imagery dataset iSAID. We also compared CFormer with other advanced methods, including ABCNet, Dcswin, Unetformer, ConvlsrNet, and UrbanSSF.

From Table IV and Fig. 5, it can be seen that CFormer outperforms all other algorithms on the iSAID dataset. In terms of mIoU, CFormer achieves improvements of 5.4%, 9.6%, 9.2%, and 9.6% over UrbanSSF, ConvlsrNet, Unetformer, and ABCNet, respectively.

E. LoveDA Dataset

To further verify the domain adaptation and generalization performance of CFormer across urban and rural environments, we conducted experiments on the LoveDA dataset, which contains diverse urban–rural scenes from three Chinese cities. We also compared CFormer with other advanced algorithms on LoveDA, including ABCNet, Dcswin, Unetformer, ConvlsrNet, SegNeXt, AFENet, and UrbanSSF.

From Table V and Fig. 6, it can be seen that the CFormer algorithm outperforms all other methods in the quantitative evaluation on the LoveDA dataset. In terms of mIoU, CFormer achieves improvements of 4.1%, 0.9%, 3.2%, and 5.4% over UrbanSSF, ConvlsrNet, Unetformer, and ABCNet, respectively.

F. WHU-OPT-SAR Dataset

To further evaluate the performance of CFormer for heterogeneous data fusion, we carried out experiments on the WHU-OPT-SAR dataset, which contains paired optical and SAR imagery. We also compared CFormer with other advanced algorithms on WHU-OPT-SAR, including ABCNet,

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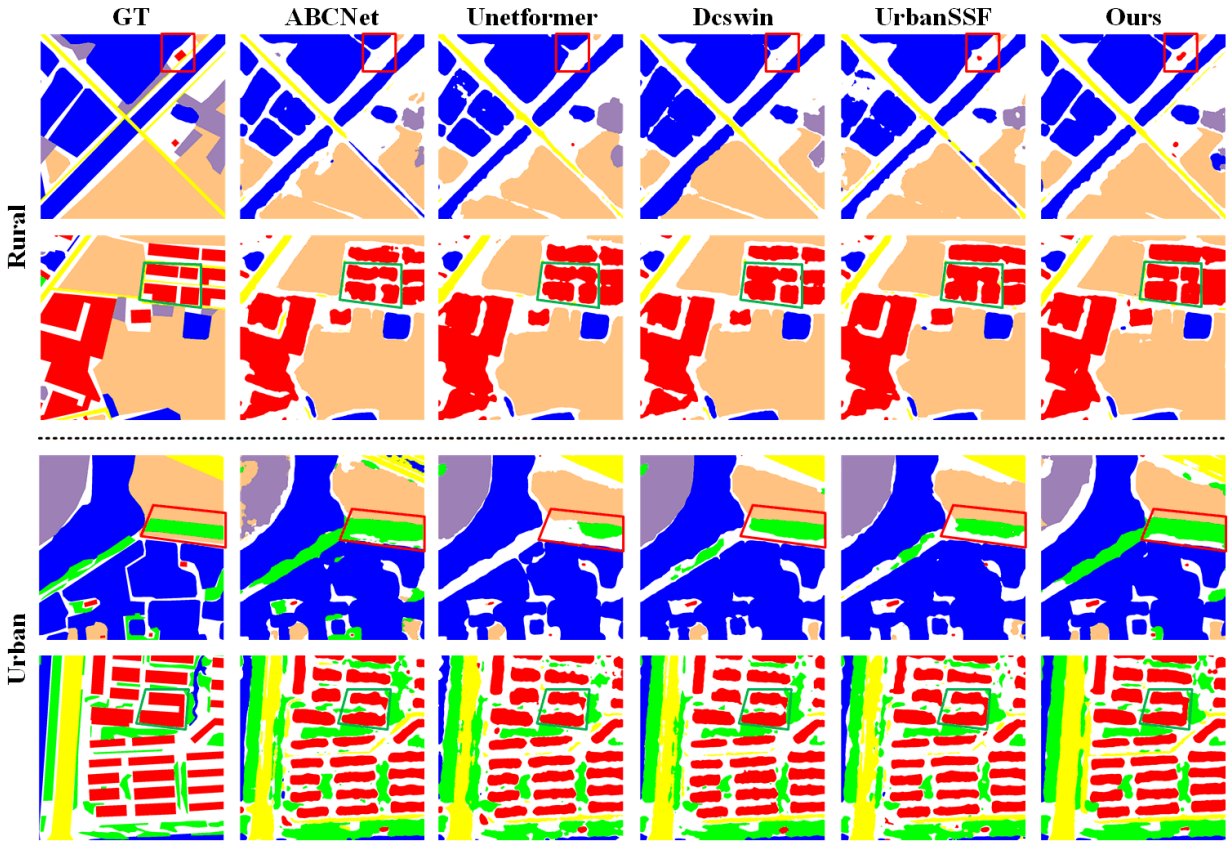


Fig. 6. Visualizations for the LoveDA dataset.

TABLE VI

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE WHU-OPT-SAR DATASET

Method	Farm.	City	Vill.	Water	Forest	Road	Oth.	mIoU	mF1	OA
ABCNet	80.1	69.9	63.0	73.0	90.6	54.8	31.8	50.4	66.1	82.4
Dcswin	80.9	71.9	64.7	76.3	90.7	57.5	37.5	55.5	68.5	83.3
Unetformer	80.5	70.0	63.9	75.6	90.5	55.7	35.6	53.1	67.4	82.9
ConvlsrNet	81.0	71.3	64.5	76.0	90.6	56.9	35.3	55.2	67.9	83.1
CSFAFormer	82.1	72.7	66.3	78.4	91.0	60.2	39.9	56.1	70.1	84.2
UrbanSSF	71.3	64.9	57.7	67.1	82.4	52.8	23.9	50.1	60.0	75.0
MIEFNet	81.1	72.1	64.9	76.7	90.9	57.4	38.9	54.7	68.9	83.5
MCKTNet	81.1	72.2	63.7	75.0	91.0	56.8	34.7	53.7	67.8	83.4
Ours	81.8	72.3	66.7	78.7	91.0	59.4	38.8	56.3	70.4	84.7

Dcswin, Unetformer, MIEFNet, MCKTNet, ConvlsrNet, CSFAFormer, and UrbanSSF.

From Table VI and Fig. 7, it can be seen that the CFormer algorithm outperforms all other methods in the quantitative evaluation on the WHU-OPT-SAR dataset. In terms of mIoU, CFormer achieves improvements of 6.2%, 1.1%, 3.2%, and 5.9% over UrbanSSF, ConvlsrNet, Unetformer, and ABCNet, respectively.

G. Ablation Study

To verify the contribution of each proposed module, we conducted an ablation study on the WHU-OPT-SAR dataset.

From Table VII, it can be observed that adding additional modules on top of the CFMST and the FRT significantly improves overall performance. The FRT constructs attention weights using complex-valued inner products in the frequency domain and restores spatial features via inverse FFT, thereby enhancing suppression of high-frequency noise and sensitivity to low-frequency structural information. This leads to greater robustness in scenes with complex backgrounds, blurred boundaries, or repetitive textures. The CFMST module extracts spatial context through multiscale convolutions and integrates a DCT-based multifrequency channel attention mechanism to primarily capture low-frequency structural information while adaptively incorporating high-frequency details for boundary refinement.

Table VIII reports the ablation results of each component in FRT. Removing the complex-valued interaction leads to

TABLE VII

ABLATION STUDY ON THE WHU-OPT-SAR DATASET (VM: VISUAL MAMBA; CFMST: CROSS-FREQUENCY MULTISCALE TRANSFORMER; FRT: FOURIER RELATIONAL TRANSFORMER; SD: STANDARD DECODER)

VM	FRT	CFMST	mIoU	mF1	OA	Params(M)	Memory(G)
Baseline Models							
SD			53.4	67.5	82.6	49.8	6.4
✓			54.5	68.5	83.1	56.4	10.0
Single Frequency Module							
SD	✓		54.9	68.9	83.8	51.6	7.0
SD		✓	55.1	69.1	83.9	59.8	9.0
Frequency Module Combinations							
SD	✓	✓	56.0	70.1	84.5	60.3	12.0
✓		✓	55.8	69.8	84.2	63.4	14.0
✓	✓		55.2	69.3	83.9	60.0	10.0
Full Model							
✓	✓	✓	**56.3**	**70.4**	**84.7**	63.8	16.0

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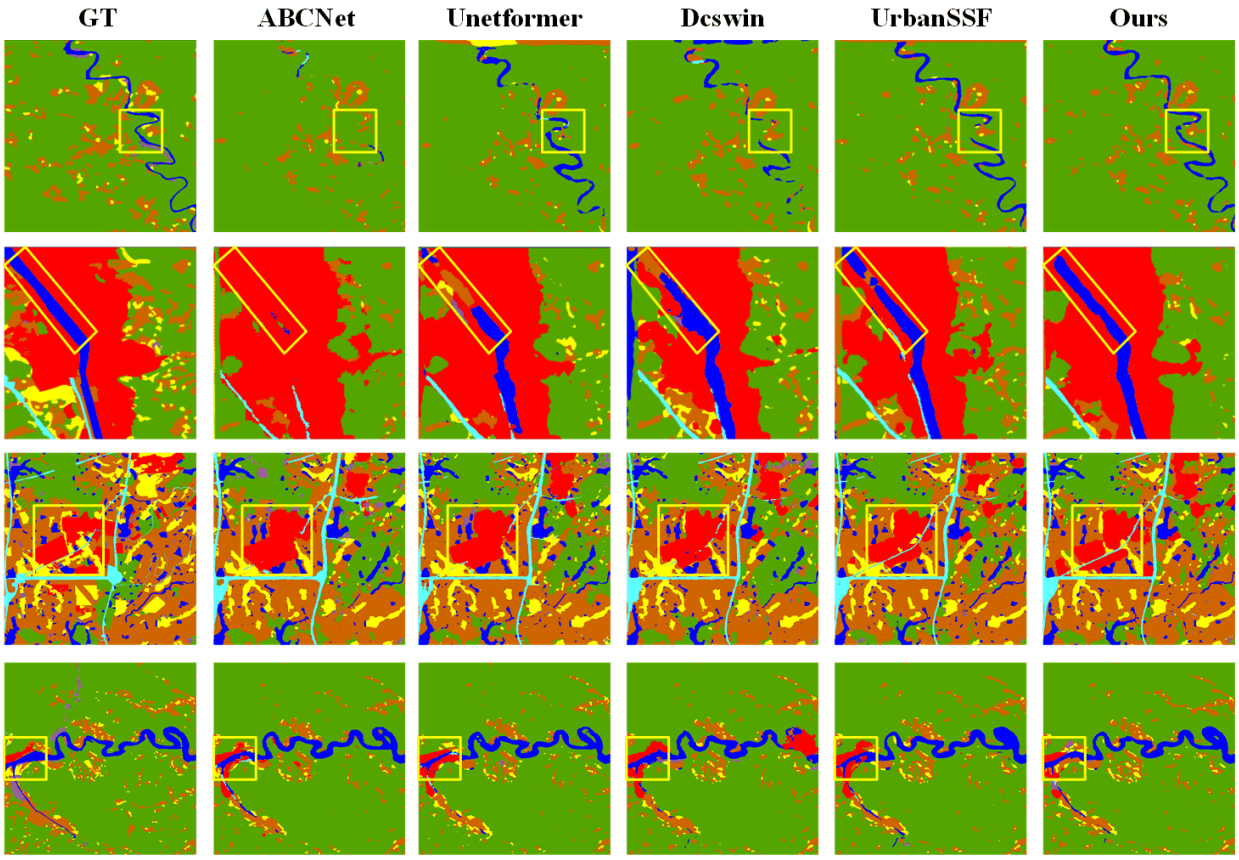


Fig. 7. Visualizations for the WHU-OPT-SAR dataset.

TABLE VIII
ABLATION STUDY OF FRT

Method	mIoU	mF1	OA
Full (FRT)	**56.3**	**70.4**	**84.7**
w/o Complex Interaction	55.3	69.3	84.0
w/o Spectral Scaling	55.2	69.2	83.8
w/o Energy Enhance	55.0	69.0	83.8
w/o Frequency Decay	54.6	68.7	83.7
w/o Post-IFFT Norm	55.4	69.4	83.9

a noticeable drop in performance, indicating that phase-aware frequency interactions are important for modeling global dependencies. Disabling spectral scaling or energy enhancement also degrades performance, suggesting that adaptive modulation of frequency responses is beneficial for segmentation. Among all components, frequency decay contributes the most significant improvement, which demonstrates its effectiveness in suppressing high-frequency noise and stabilizing frequency responses. The post-IFFT normalization mainly improves training stability and provides a modest performance gain.

H. Spectral Attribution Analysis

To further understand the behavior of FRT, we conduct radial spectral response analysis and controlled band-masking experiments on the ISPRS Vaihingen dataset.

Fig. 8 presents the radial-averaged spectral responses under different masking strategies. Masking high-frequency components results in nearly identical spectral curves compared to the baseline, indicating that high-frequency bands contribute marginally to the learned representation. In contrast, masking low-frequency components leads to a substantial suppression of spectral energy across the entire spectrum.

TABLE IX
SPECTRAL ENERGY DECOMPOSITION UNDER DIFFERENT MASKING STRATEGIES

Mode	Total Area	Low Ratio	Mid Ratio	High Ratio
None	23.24	0.394	0.202	0.366
Mask High	23.24	0.394	0.202	0.366
Mask Low	3.76	0.934	0.033	0.022

Quantitatively, as summarized in Table IX and Fig. 9, removing high-frequency components does not alter the total spectral energy (23.24), whereas removing low-frequency components reduces the total spectral energy by 83.8%.

The energy decomposition further shows that approximately 39.4% of the spectral energy lies within the low-frequency band ($r \leq 0.25$), while mid- and high-frequency components account for 20.2% and 36.6%, respectively. Although the absolute proportion of low-frequency energy is not overwhelmingly dominant, its removal causes catastrophic energy loss, demonstrating that semantic segmentation is predominantly governed by structural consistency, while fine-grained texture responses play a complementary role.

To assess whether boundary prediction is predominantly driven by high-frequency components, we evaluate the boundary-$F1$ (B$F1$) score under different pixel tolerances. B$F1$ increases monotonically with tolerance ($t = 1 \rightarrow 0.387$,

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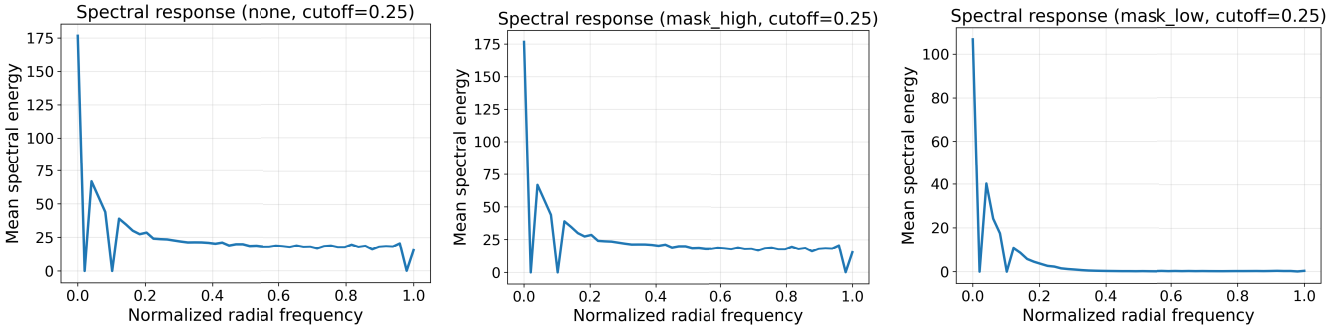


Fig. 8. Radial spectral response curves under different band-masking settings. Removing high-frequency components does not alter the spectral profile, while removing low-frequency components drastically suppresses total spectral energy.

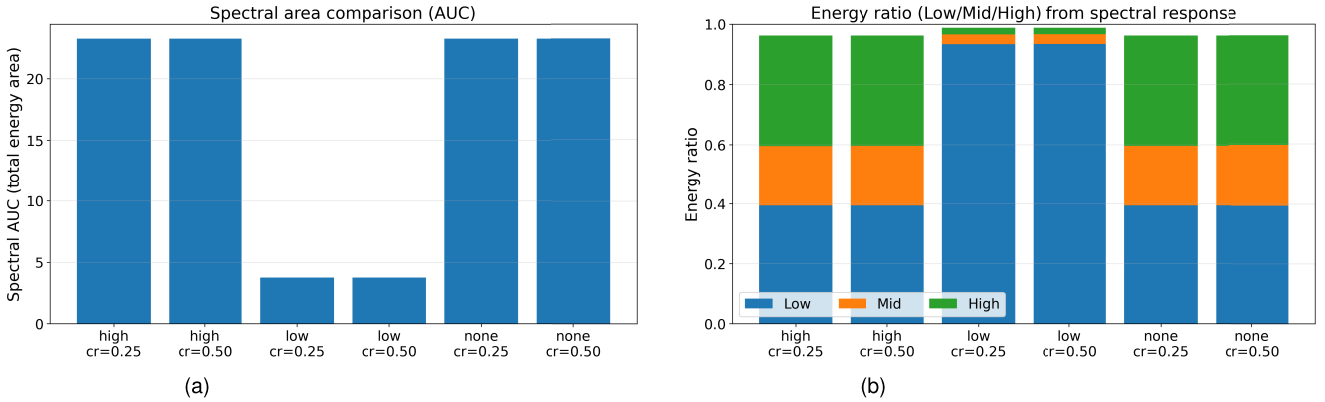


Fig. 9. Quantitative spectral analysis. (a) Removing high-frequency components does not change total energy, while removing low-frequency components reduces energy by over 80. (b) Frequency band decomposition shows that structural low-frequency components dominate the model representation.

$t = 2 \rightarrow 0.600$, and $t = 3 \rightarrow 0.720$) due to relaxed matching criteria. Importantly, no notable differences are observed across masking settings, indicating that boundary prediction is not predominantly determined by high-frequency components alone.

Overall, these results provide quantitative spectral evidence that the proposed frequency-domain modeling captures meaningful structural representations beyond conventional spatial attention mechanisms.

V. Conclusion

This article proposes a remote sensing semantic segmentation method based on a cross-frequency transformer, aiming to improve segmentation performance under multiscale object variation and complex texture conditions. CFMST captures large-scale structural variations through multiscale convolutions and employs a multifrequency channel attention mechanism with dominant reliance on low-frequency global semantics and supplementary high-frequency cues for local refinement, enhancing the network's ability to analyze boundaries, textures, and complex backgrounds. FRT transfers attention computation to the frequency domain, enabling more compact modeling of long-range dependencies while improving robustness to noise and mixed pixels.

Although frequency-domain modeling enhances structural consistency and robustness, it may introduce over-smoothing in extreme cases involving very small or low-contrast objects. To alleviate this issue, the proposed framework integrates multifrequency channel attention and multiscale refinement to preserve high-frequency boundary details. Nevertheless, adaptive frequency selection or dynamic gating mechanisms could further improve the balance between global structure modeling and fine-detail retention, which will be explored in future work.

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